

## Lecture 5

### Higher Dimensional eq.

$$v: \mathbb{R}_t \times \mathbb{R}_x^n \rightarrow \mathbb{R}^n, \quad w: \mathbb{R} \times \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}.$$

Suppose  $u$  solves

$$\frac{d}{dt} u(t, x) + v(t, x) \cdot \nabla u(t, x) + w(t, x, u(t, x)) = 0$$

Def.  $x(t)$  a characteristic if

$$\frac{dx}{dt}(t) = v(t, x(t)).$$

Thm. If  $x(t)$  is a characteristic, then

$$\frac{d}{dt} u(t, x(t)) + w(t, x(t), u(t, x(t))) = 0$$

Ex.  $x \in \mathbb{R}^2$ .  $\partial_t u + \partial_{x_1} u + \partial_{x_2} u = 0$ ,  $u(0, x) = f(s)$

$$\frac{d}{dt} x(t) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$x(t, s) = \begin{pmatrix} t \\ t \end{pmatrix} + s \quad s \in \mathbb{R}^2$$

$$u(t, t+s_1, t+s_2) = f(s)$$

$$u(t, x) = f(x_1 - t, x_2 - t)$$

## Quasilinear 1st order PDEs

Consider  $u \in C^1([0, T] \times \Omega)$ ,  $\Omega \subset \mathbb{R}$ , satisfying

$$\textcircled{*} \begin{cases} \partial_t u + a(u) \partial_x u = 0 \\ u(0, x) = f(x) \end{cases}$$

where  $a \in C^1(\mathbb{R})$  and  $f \in C^1(\Omega)$

Def.  $x(t)$  is a characteristic of  $\textcircled{*}$  if  
 $x(t) = x(t, s) = s + a(f(s))t$ ,  $s \in \Omega$

Thm. Suppose  $u \in C^1([0, T] \times \Omega)$  solves  $\textcircled{*}$ , then

$\forall s \in \Omega$ ,  $u$  is constant along the characteristics of  $\textcircled{*}$ .

Pf. Let  $x(t)$  be a characteristic. Then note that

$$u(0, x(0)) = f(s)$$

$$\begin{aligned} \text{and } \frac{d}{dt} u(t, x(t)) &= \partial_t u(t, x(t)) + \partial_x u(t, x(t)) \partial_t x(t) \\ &= -a(u(t, x(t))) \partial_x u(t, x(t)) \\ &\quad + a(f(s)) \partial_x u(t, x(t)) \end{aligned} \quad \left. \vphantom{\frac{d}{dt} u(t, x(t))} \right\} \textcircled{**}$$

Observe that  $u(t, x(t)) = f(s)$  is a solution to  $\textcircled{**}$ ,  
hence it is the only solution  $\square$

Rmk. Unlike in the linear case, the characteristics in the quasilinear case depends on initial condition  $u(0, x)$ .

In the linear case, by existence and uniqueness of ODEs, characteristic lines cannot intersect one another, we will see this is no longer true in quasilinear case.

Ex. (Burger's eq)  $u: \mathbb{R}_t \times \mathbb{R}_x \rightarrow \mathbb{R}$

$$\partial_t u + u \partial_x u = 0 \quad (\text{BE})$$

Experiment 1 (BE) with initial cond is

$$u(0, x) = ax + b, \quad a, b \in \mathbb{R}.$$

Characteristics  $x(t, s) = s + t(as + b)$ .

solution  $u(t, x(t, s)) = as + b$

$$\Rightarrow u(t, s + t(as + b)) = as + b$$

note  $s = \frac{x - tb}{1 + ta}$

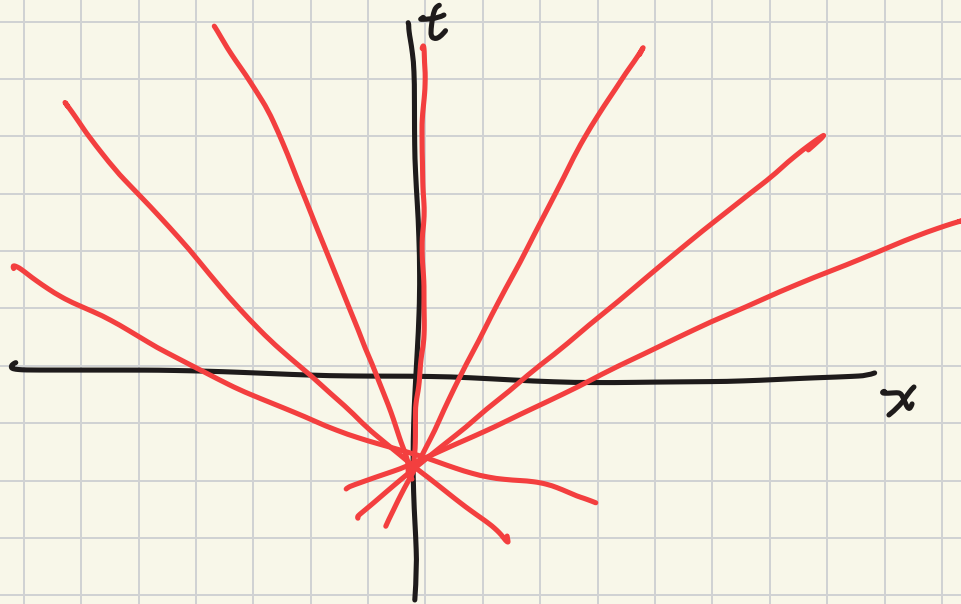
$$\begin{aligned} \Rightarrow u(t, x) &= a \frac{x - tb}{1 + ta} + b \\ &= \frac{ax + b}{1 + ta} \end{aligned}$$

## Observations

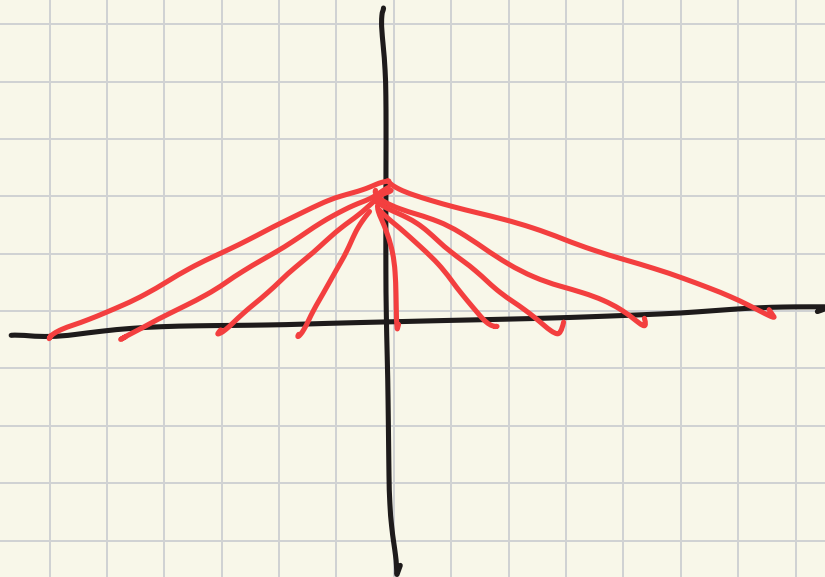
1.) If  $a > 0$ , then solution exists  $\forall t > 0$

2.) If  $a < 0$ , then solution exists when  $t < -\frac{1}{a}$

$$a > 0$$
$$b = 0$$



$$a < 0$$
$$b = 0$$



Characteristics intersect at  $t = -\frac{1}{a} !!$

Experiment 2 (BE) w/ arbitrary initial cond is

$$u(0, x) = f(x).$$

Then if  $u$  is a solution, it satisfies

$$u(t, s + t f(s)) = f(s)$$

To solve (BE), we must solve

$$x = s + t f(s)$$

for  $s$ . Cannot do this explicitly for most  $f$ , but we have the following

Thm (Implicit Fcn Thm). Let  $f: \mathbb{R}_x^n \times \mathbb{R}_y^m \rightarrow \mathbb{R}^m$  be continuously differentiable. Suppose

$$f(x_0, y_0) = 0$$

and  $\det(D_y f(x_0, y_0)) \neq 0$ , then  $\exists$  nbhds  $U$  and  $V$  of  $x_0$  and  $y_0$  respectively and  $g: U \rightarrow V$  s.t. it is continuously differentiable and

$$f(x, g(x)) = 0 \quad \forall x \in U.$$

Applied to our setting, define

$$F(t, x, s) = s + t f(s) - x = 0$$

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$$F(t, x, s) = s + t f(s) - x = 0$$

fix  $(t, x, s) = (0, x_0, x_0)$ , observe

$$\frac{\partial}{\partial s} F(0, x_0, x_0) = 1 + 0 \cdot f(x_0) = 1$$

$\Rightarrow$  always exists  $s = s(x, t)$  s.t.  $s + t f(s) = x$

$\Rightarrow u(t, x) = f(s(x, t))$  is a solution to  
(BE) w/ initial condis  $u(0, x) = f(x)$  for  $t$  small

Telegraphers Eq.

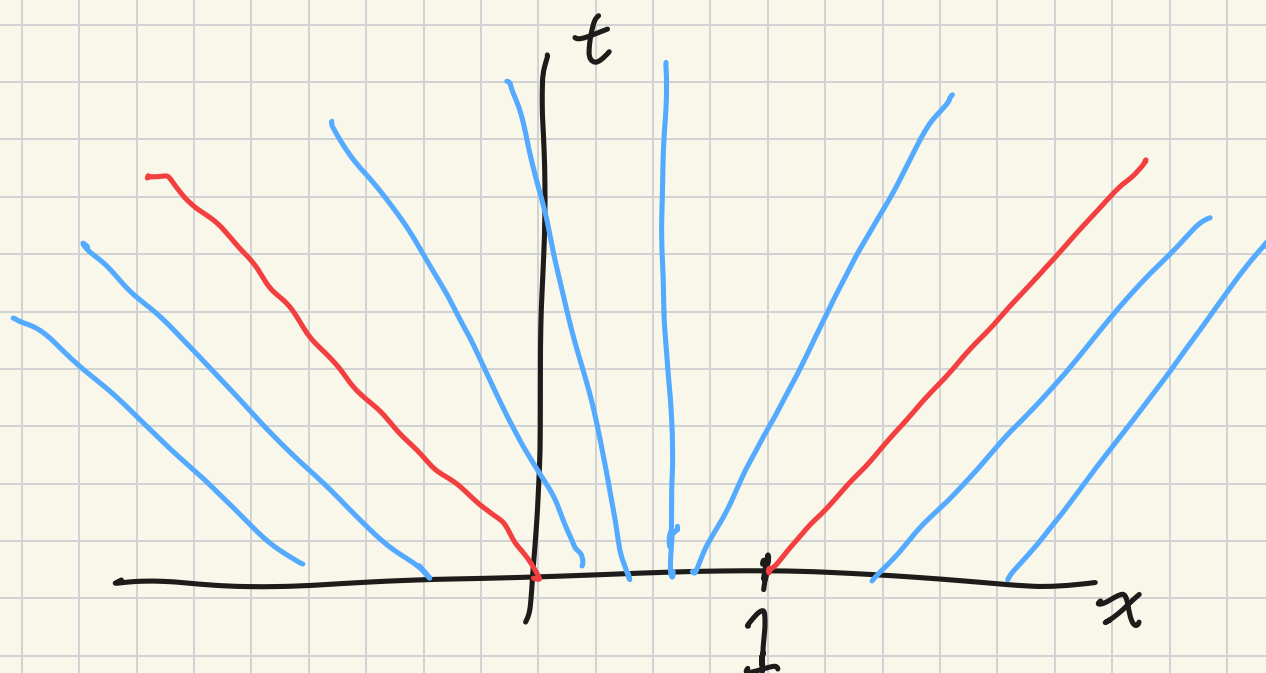
$$\frac{\partial u}{\partial t} + (1 - 2u) \frac{\partial u}{\partial x} = 0$$

$$u(0, x) = f(x)$$

with

$$f(x) = \begin{cases} 1 & x \leq 0 \\ 1-x & 0 \leq x \leq 1 \\ 0 & x \geq 1 \end{cases}$$

$$\text{Characteristics } x(t, s) = \begin{cases} s - t & x(t, s) \leq -t \\ s + t(2x(0) - 1) & -t \leq x(t, s) \leq 1 + t \\ s + t & x(t, s) \geq 1 + t \end{cases}$$



$$\text{for } x \leq -t \quad s = x + t$$

$$\text{for } -t \leq x \leq t+1 \quad s = \frac{x+t}{1+2t}$$

$$\text{for } x \geq 1+t \quad s = x - t$$

$$\Rightarrow u(t, x) = \begin{cases} 1 & x \leq -t \\ \frac{-x+t-1}{1+2t} & -t \leq x \leq t+1 \\ 0 & x \geq 1+t \end{cases}$$

Relationship between Burger and Telegrapher

Suppose  $u$  solves Telegrapher, put

$$v(t, x) = 1 - 2u(t, x)$$

Then  $v(t, x)$  solves Burger's eq.

Additional into.

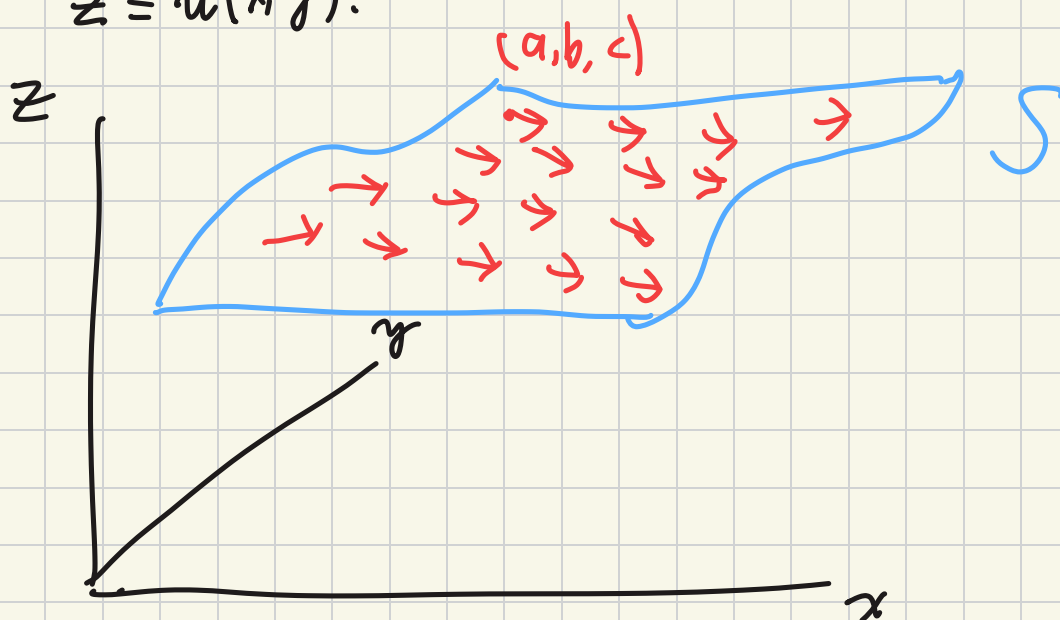
Some perspective on general quasilinear

1st order PDEs

$u: \mathbb{R}^2 \rightarrow \mathbb{R}$  satisfies,  $u(x, y)$

$$\textcircled{*} \quad a(x, y, u) \partial_x u + b(x, y, u) \partial_y u = c(x, y, u)$$

Put  $z = u(x, y)$ .



$(\partial_x u, \partial_y u, -1)$  is normal to  $S$

$\Rightarrow (a(x, y, z), b(x, y, z), c(x, y, z))$  tangent to  $S$ .

Def  $(x, y, z)(t)$  is a characteristic if

$$\frac{d}{dt} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a(x, y, z) \\ b(x, y, z) \\ c(x, y, z) \end{pmatrix}$$

Thm. If  $(x_0, y_0, z_0)$  lies on  $S$ , then the characteristic through  $(x_0, y_0, z_0)$  lies on  $S$